Lecture 1 Slides Math Modelling and ODEs

Lecture 1
Slides Ma...

Engineering Mathematics III

ENGE702

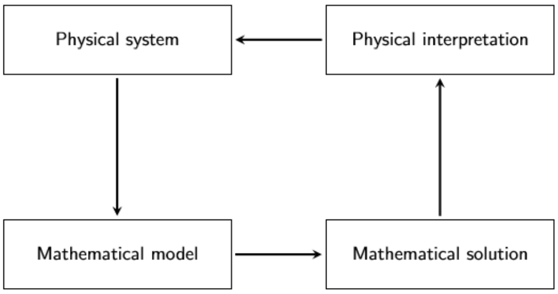
Lecture 1: Mathematical modelling and ODEs

Modelling

- A mathematical model attempts to describe a physical process or phenomenon mathematically
- The resulting model is solved mathematically, if possible, and the solution interpreted physically.
- The process of modelling is something of an art. A system does not have a unique 'correct' model, but some models may be more correct (and more useful) than others.
- The aim is to translate the underlying physical principles to mathematical language.
- Many engineering processes can be modelled with differential equations. This is what we will focus on in this course.

The modelling process

Modelling, solving, interpreting, possibly remodelling



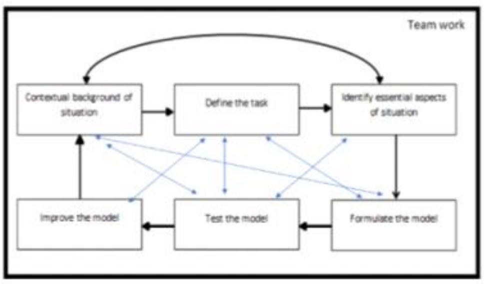
Modelling

Some examples:

A small heavy ball is dropped from a tall building. How could we model the position of the ball as a function of time?

We will return to this later in next lecture.

Modelling steps



Modelling steps

For a modelling project, here are key steps

- Establish a shared understanding for the problem situation.
- Define the problem.
- Recognise and identify the essential aspects of the situation.
- Formulate a model.
- Test the model
- Improve the model
- Reporting on the model.

Modelling tips

The KISS principle "Keep It Simple Stupid"

- Start by developing a simple model
- Then extend and improve the model by incorporating further features
- In practice models are very simplified and often only attempt to model part of a system.
- Always start by considering the simplest model, and then add in more complexities to make the model more realistic.
- Declare any assumptions made and consider these when making interpretations of results produce by the model.

Modelling

Some examples:

How could we model the velocity of a parachutist dropped from an aeroplane?

Once the parachute opens, air resistance must be taken ino account.

This is discussed in the textbook. See Problem Set 1.2, Q15.

Modelling

Some examples:

How could we model the displacement of a mass on a spring?

We will return to this problem in detail in later lectures.

More modelling examples

- Modelling water levels in tanks with inflow and outflow,
- beats of a vibrating system,
- current in an electrical circuit,
- deformation of a beam,
- simple harmonic motion (a swinging pendulum),
- earthquake shaking of multistory buildings (spring system),
- Lotka-Volterra predator prey models (environmental applications),
- epidemiology (Covid modelling) and
- modelling physiology (for example, diabetes).

See the textbook, Section 1.1, p3.

The last three examples are not directly from engineering but give a glimpse of how widely differential equations are applied. The skills you develop in this course will be very useful in engineering and also in many other branches of science. You may find yourself collaborating with scientists from other disciplines in your engineering career.

Differential Equations

A differential equation is a equation that involves derivatives.

Differential equations that involve functions of only one variable are called **ordinary differential equations (ODEs)**. We will study these for the first half of the course.

Examples:

- $y'' + 3y' = 2t^2$
- $\sin(t)y''' - 2yy' + 3ty = 0$
- $\dfrac{d^2y}{dt^2} + 5t\dfrac{dy}{dt} = \cos(2t)$

y dependent variable

t independent variable

$y'' = \dfrac{d^2y}{dt^2}$

Here the notation y' refers to the first derivative of y, when y is a function of one variable (usually t or x). The notation y'' refers to the second derivative of y. We also use the Leibniz notation $\dfrac{dy}{dt}$ or $\dfrac{dy}{dx}$ in place of y', and $\dfrac{dy^2}{dt^2}$ in place of y'' etc.

Algebraic equation:

$\frac{1}{2}x \quad 5 = 0 \Rightarrow x = 10$

$x^2 + 2x + 3 = 0 \Rightarrow (x+1)(x+2) = 0$

$\Rightarrow x = -1$

or $x = -2$

Differential equation is an equation involving derivative

Example: $y' = x$

$\dfrac{dy}{dt} + t^2 = 0$,

$\dfrac{d^2y}{dt^2} + \dfrac{dy}{dt} + e^t = 0$

only one independent variable in the derivatives

ODEs

$\dfrac{\partial y}{\partial t} + \dfrac{\partial y}{\partial x} = 0$

more than one independent variables in the derivatives: PDEs

Differential Equations

Differential equations that involve functions of two or more variables are called **partial differential equations (PDEs)**. These will be covered in the second half of the course.

Examples:

Differential Equations

Differential equations that involve functions of two or more variables are called **partial differential equations (PDEs)**. These will be covered in the second half of the course.

Examples:

- $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$ one dimensional wave equation.
- $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$ one dimensional heat equation.
- $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ two dimensional Laplace equation.
- $\frac{\partial^2 u}{\partial t^2} = c^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right)$ two dimensional wave equation.
- $\frac{\partial u}{\partial t} = c^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}\right)$ three dimensional heat equation.

only one independent variable in the derivatives ODEs

derivatives: PDEs

Differential Equations

dependent variable is a function of independent variable

A **solution** to a differential equation is a function that 'satisfies' the equation or makes the equation true. For example:

$$y = cx^2$$

is a solution to the differential equation

$$xy' = 2y$$

We can see this since $y' = 2cx$, so $xy' = 2cx^2$. Also $2y = 2cx^2$, so both sides are the same.

Example: Verify $y = cx^2$ is a solution to the ODE $xy' = 2y$.

For the ordinary differential equation (ODE)

$LHS = xy' = x \cdot (2cx) = 2cx^2.$

$RHS = 2y = 2 \cdot cx^2 = 2cx^2$

So $LHS = RHS$, $y = cx^2$ satisfies the ODE $xy' = 2y$.

Differential Equations

Show

$$y = ce^x$$

is a solution to the differential equation

$$y'' - y = 0$$

Since $y = ce^x$, we find $y' = ce^x$ and also $y'' = ce^x$, hence $y'' - y = 0$.

Example.

Show $y = ce^x$ is a solution to ODE $y'' - y = 0$

$LHS = y'' - y = (ce^x)'' - ce^x$
$= ce^x - ce^x = 0$

$RHS = 0$

So $y = ce^x$ satisfies the ODE $y'' - y = 0$.

Differential Equations

Show

$$y = c_1 \sin(x) + c_2 \cos(x)$$

$LHS = y'' + y$

Show

$$y = c_1 \sin(x) + c_2 \cos(x)$$

is a solution to the differential equation

$$y'' + y = 0$$

Since $y = c_1 \sin(x) + c_2 \cos(x)$, we find $y' = c_1 \cos(x) - c_2 \sin(x)$ and $y'' = -c_1 \sin(x) - c_2 \cos(x)$, hence $y'' + y = 0$.

$LHS = y'' + y$

$= (c_1 \sin(x) + c_2 \cos(x))'' + c_1 \sin(x) + c_2 \cos(x)$

$= (c_1 \cos(x) - c_2 \sin(x))' + c_1 \sin(x) + c_2 \cos(x)$

$= -c_1 \sin(x) - c_2 \cos(x) + c_1 \sin(x) + c_2 \cos(x)$

$= 0 = RHS$

Differential Equations

Exercises
Show the following functions are solutions to the corresponding differential equations:

1. $y = ce^{kx}$ and $y' = ky$,
2. $y = c_1 \sin(2x) + c_2 \cos(2x)$ and $y'' + 4y = 0$.

Exercises are discussed in our **weekly joint problem-solving sessions**.

Don't forget to do Quiz 1. It is based on the material for the week and is open for a few days. The due date will be visible on Canvas. **These quizzes are graded.** The grades form part of the assignment grades.

Text Book References

Chapter 1.1 of Advanced Engineering Mathematics by E. Kreyszig.